

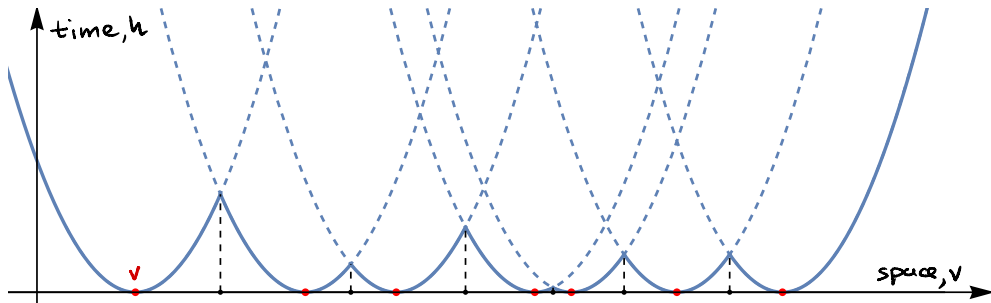
Gaussian Approximation in Delaunay Tessellations - Counting Extreme Vertices via Region Stabilization

Joint with Anna Gusakova (University of Münster)

Chinmoy Bhattacharjee

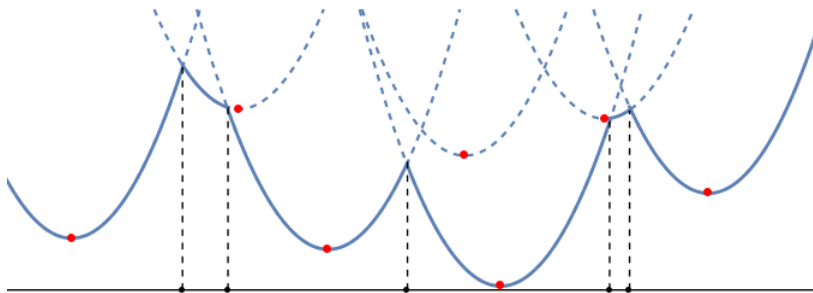
27.10.2025

Voronoi $\rightarrow$ Laguerre Tessellation



Paraboloid growth process in a Voronoi tessellation

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Paraboloid growth process in a [Laguerre](#) tessellation

- Let η be a Poisson point process in space-time $\mathbb{X} := \mathbb{R}^d \times E$, $E \subseteq \mathbb{R}$, with intensity measure $\mathbb{Q} = \lambda \otimes F$, where λ is the Lebesgue measure and F is a locally finite Borel measure.

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- The Laguerre tessellation is the collection $\mathcal{L}(\eta) = \{C((v, h), \eta) : (v, h) \in \eta\}$, where we define the **cell** associated to a nucleus $x = (v, h) \in \eta$ as

$$C((v, h), \eta) := \{w \in \mathbb{R}^d : \|w - v\|^2 + h \leq \|w - v'\|^2 + h' \quad \forall (v', h') \in \eta\}.$$

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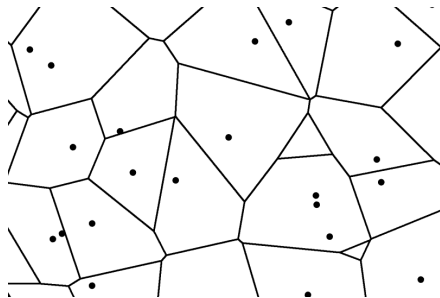
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- If all $h = 0$, then specializes to classical Voronoi tessellation.
- Unlike Voronoi, it is not necessary that a cell is non-empty, or that it contains its nucleus.

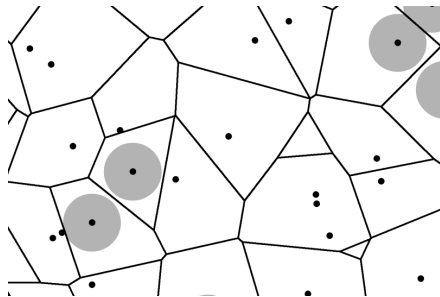
Crystal Growth

Each seed in η after being activated, starts forming a *crystal/cell* by growing isotropically with the same decaying speed $1/\sqrt{t}$ after time $t > 0$ of being active. A point $w \in \mathbb{R}^d$ is included in the cell of $(v, h) \in \eta$ if its crystal reaches w first.



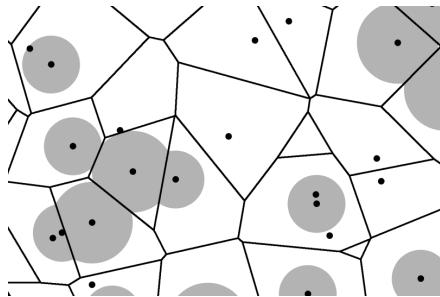
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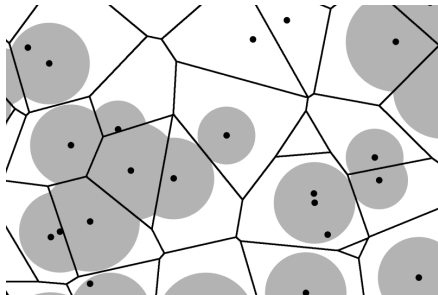
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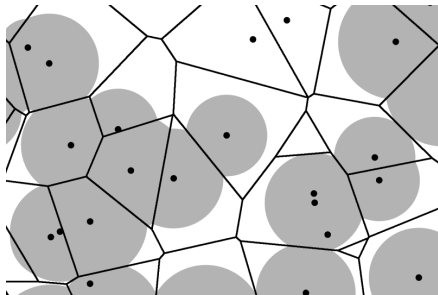
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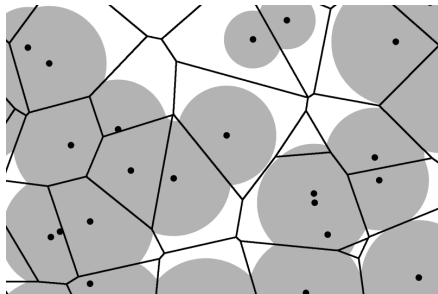
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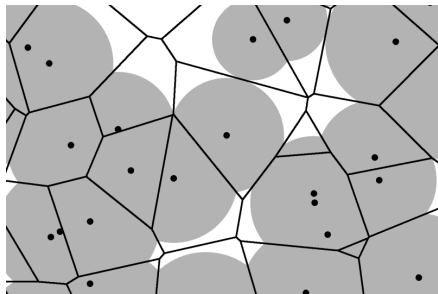
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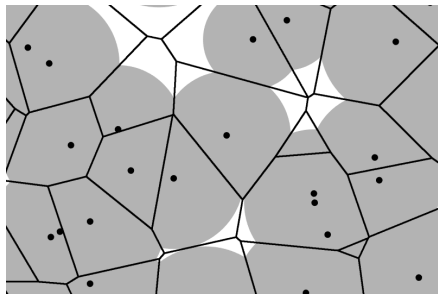
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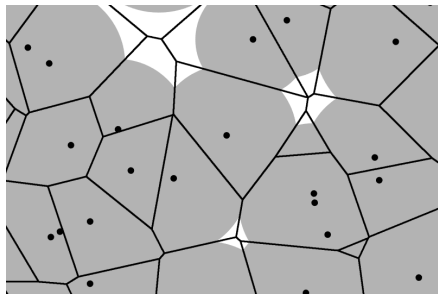
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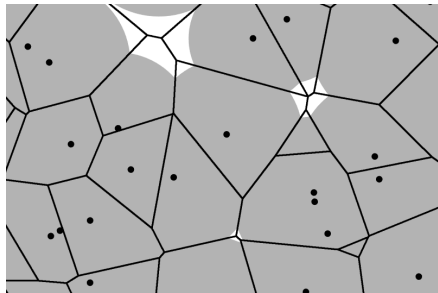
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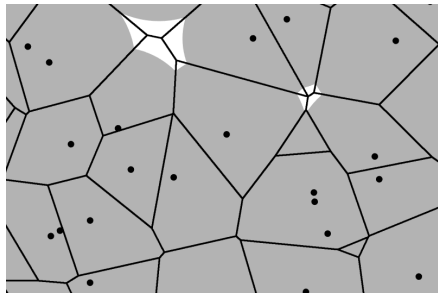
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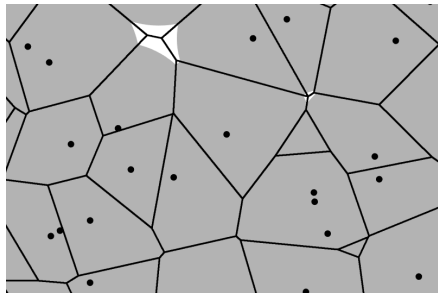
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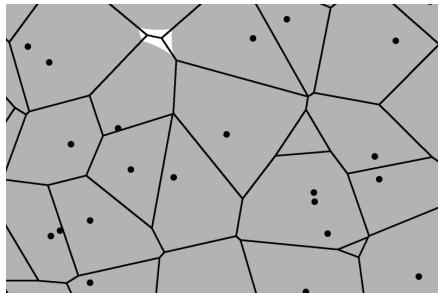
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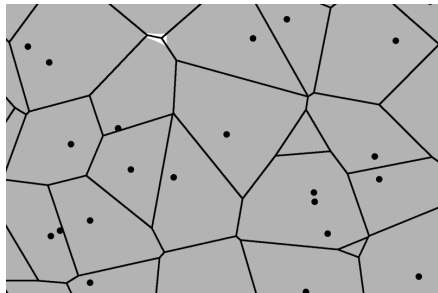
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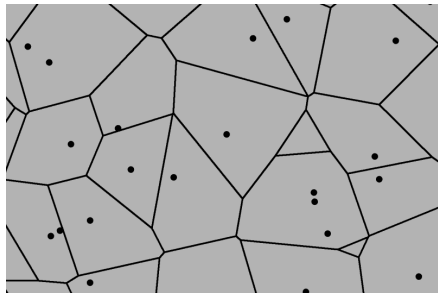
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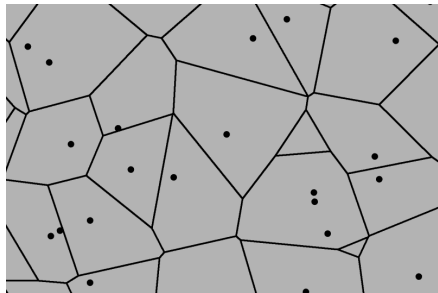
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Applications in cell biology, fluid mechanics, and materials science, to name a few.

Motivation - Sectional Voronoi Tessellation

- Given a Voronoi tessellation $\mathcal{V}_d$ and an ℓ -dimensional affine subspace $L_\ell \subset \mathbb{R}^d$, define the sectional Poisson-Voronoi tessellation as

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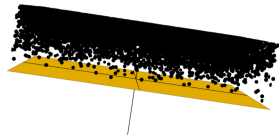
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- Indeed, they form the so-called β -Voronoi tessellation.

Choice of Poisson Processes

β -Voronoi : For $\beta \in (-1, \infty)$, let η_β denote a Poisson point process in $\mathbb{R}^d \times [0, \infty)$ with intensity measure $\mathbb{Q}_\beta$ given by $h^\beta dh dv$.

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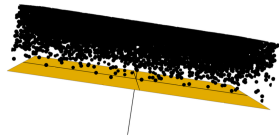
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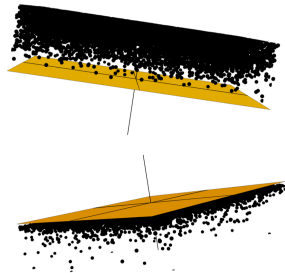
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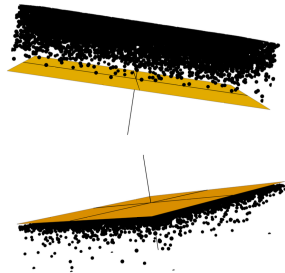


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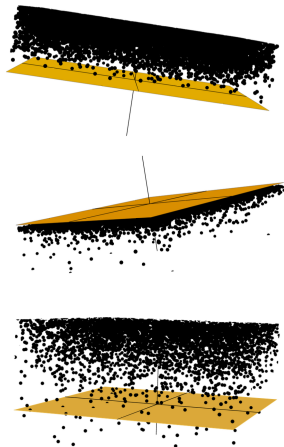


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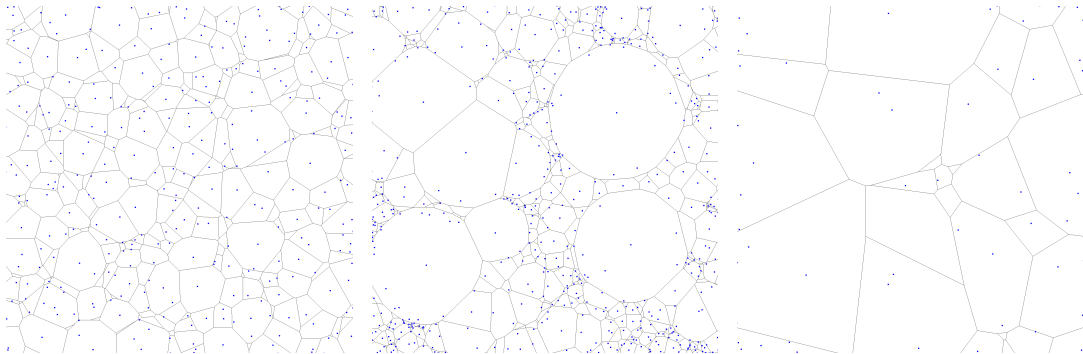
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Different Behaviour



Left: β -Voronoi tessellation in $\mathbb{R}^2$ with $\beta = 5$. Middle: β' -Voronoi tessellation in $\mathbb{R}^2$ with $\beta = 2.5$. Right: Gaussian-Voronoi tessellation in $\mathbb{R}^2$. The corresponding centers of growth are marked in blue.

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For $W_n := [-n, n]^d$, $n \in \mathbb{N}$, and $\eta \in \{\eta_\beta, \eta'_\beta, \tilde{\eta}\}$, let $F_n(\eta)$ denote the **number of extreme points** with their nucleus lying in the window W_n , given by

$$F_n(\eta) := \sum_{(v, h) \in \eta} \mathbb{1}(v \in W_n) \mathbb{1}((v, h) \text{ is extreme in } \mathcal{L}(\eta)).$$

Statistic of interest

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Note here that the associated cells themselves might possibly lie outside of the window W_n , partly or even wholly.

Earlier Literature

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Variance lower (Gusakova et. al. '23, B., Gusakova '25+)

We have

$$\text{Var}(F_n(\eta_\beta)) \geq C_1 n^d, \quad \text{Var}(F_n(\eta'_\beta)) \geq C_2 n^d, \quad \text{Var}(F_n(\tilde{\eta})) \geq C_3 n^d$$

for constants $C_1, C_2, C_3 \in (0, \infty)$ depending only on d and β .

Main Result: Quantitative CLT for Extreme Points

Quantitative CLT for Extreme Points (B., Gusakova '25+)

Let η be either (a) η_β with $\beta > -1$, or (b) η'_β with $\beta > 5d + 1$, or (c) $\tilde{\eta}$. Then there exists $0 < C_1 \leq C_2 < \infty$ such that

$$C_1 n^d \leq \text{Var}(F_n(\eta)) \leq C_2 n^d.$$

Moreover, for $N \sim \mathcal{N}(0, 1)$, there exists a constant $C \in (0, \infty)$ depending only on d, β such that for all $n \in \mathbb{N}$,

$$d\left(\frac{F_n(\eta) - \mathbb{E}F_n(\eta)}{\sqrt{\text{Var } F_n(\eta)}}, N\right) \leq \frac{C}{n^{d/2}}$$

for $d \in \{d_W, d_K\}$.

Representation as a Sum of Scores

Let $\mathbf{N}$ be the family of σ -finite counting measures $\mathcal{M}$ on $\mathbb{X} = \mathbb{R}^d \times \mathbb{R}$. We can represent

$$F_n(\eta) = \sum_{x \in \eta} \xi_n(x, \eta),$$

where the score function $\xi_n : \mathbb{X} \times \mathbf{N} \rightarrow \mathbb{R}$ is given by

$$\xi_n((v, h), \mathcal{M}) := \mathbb{1}(v \in W_n) \mathbb{1}((v, h) \text{ is an extreme point in } \mathcal{M}), \quad (v, h) \in \mathcal{M} \in \mathbf{N}.$$

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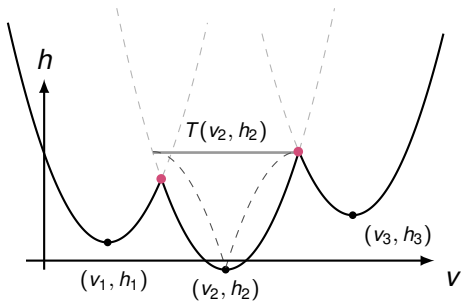
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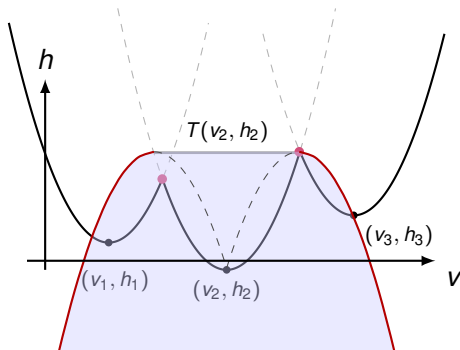
The widely applicable approach via exponentially decaying radius of stabilization due to [Lachièze-rey, Schulte and Yukich '19](#) doesn't seem to apply here.

Not Exponentially Stabilizing in the sense of LSY '19



T is the coverage time.

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T is the coverage time. Any point added in the blue **unbounded** downward paraboloid may affect the extremal property of the point (v_2, h_2)

Region stabilization - Assumptions

For $(\mathbb{X}, \mathcal{F})$ Borel and η a Poisson process with intensity $\mathbb{Q}$ where $\mathbb{Q}$ is σ -finite, consider a sum of the form $H_n = \sum_{x \in \eta} \xi_n(x, \eta)$.

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(A1) Stabilization region: For all $n \geq 1$, there exists a **set-valued** map $R_n : \mathbb{X} \times \mathbf{N} \rightarrow \mathcal{F}$ such that for $x \in \mathcal{M}$,

$$\xi_n(x, \mathcal{M}) = \xi_n(x, \mathcal{M} \cap R_n(x, \mathcal{M}));$$

+ some natural conditions on ξ_n and R_n + measurability.

Let $r_n : \mathbb{X} \times \mathbb{X} \rightarrow [0, \infty]$ be a non-zero function such that

$$\mathbf{P}\{y \in R_n(x, \eta + \delta_x)\} \leq e^{-r_n(x,y)}, \quad x, y \in \mathbb{X}.$$

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(A2) There exists $p > 0$ such that, for $\mathcal{M} \in \mathbf{N}$ with $\mathcal{M}(\mathbb{X}) \leq 7$,

$$\left\| \xi_n(x, \eta + \delta_x + \mathcal{M}) \right\|_{4+p} \leq M_{n,p}(x), \quad s \geq 1, \quad x \in \mathbb{X},$$

Quantitative CLT via region stabilization (B. and Molchanov '22)

Assuming **(A1)**, **(A2)** and writing $\mathbb{Q}f := \int f(x) \mathbb{Q}(\mathrm{d}x)$,

$$\mathrm{Var}(H_n) \leq C \mathbb{Q} \left((\kappa_n + g_n)^{2\beta} G_n \right),$$

$$d_W \left(\frac{H_n - \mathbb{E}H_n}{\sqrt{\mathrm{Var} H_n}}, N \right) \leq C \left[\frac{\sqrt{\mathbb{Q}f_\beta^2}}{\mathrm{Var} H_n} + \frac{\mathbb{Q}((\kappa_n + g_n)^{2\beta} G_n)}{(\mathrm{Var} H_n)^{3/2}} \right];$$

and a similar bound holds for the Kolmogorov distance.

Quantitative CLT via region stabilization (B. and Molchanov '22)

Assuming **(A1)**, **(A2)** and writing $\mathbb{Q}f := \int f(x) \mathbb{Q}(dx)$,

$$\begin{aligned}\mathrm{Var}(H_n) &\leq C \mathbb{Q} \left((\kappa_n + g_n)^{2\beta} G_n \right), \\ d_W \left(\frac{H_n - \mathbb{E} H_n}{\sqrt{\mathrm{Var} H_n}}, N \right) &\leq C \left[\frac{\sqrt{\mathbb{Q} f_\beta^2}}{\sqrt{\mathrm{Var} H_n}} + \frac{\mathbb{Q}((\kappa_n + g_n)^{2\beta} G_n)}{(\mathrm{Var} H_n)^{3/2}} \right];\end{aligned}$$

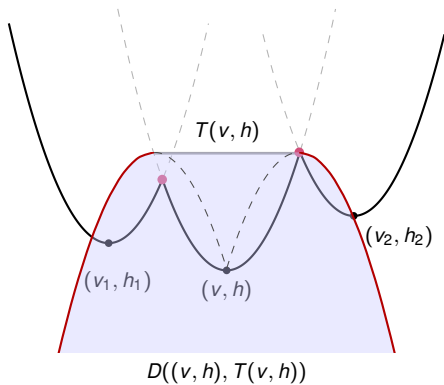
and a similar bound holds for the Kolmogorov distance. Here

$$\kappa_n(x) = \mathbf{P} \{ \xi_n(x, \eta + \delta_x) \neq 0 \}$$

and C, β are explicit constants depending on p . The functions f_β , g_n and G_n are certain integrals involving

$$\mathbf{P}\{y \in R_n(x, \eta + \delta_x)\} \quad \text{and} \quad \mathbf{P}\left\{\{y_1, y_2\} \subseteq R_n(x, \eta + \delta_x)\right\}.$$

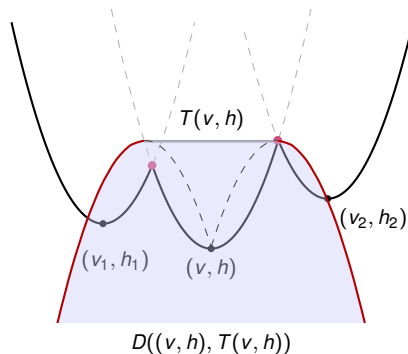
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$$R_n((v, h), \eta + \delta_{(v, h)}) \\ := \begin{cases} D((v, h), T((v, h))), & \text{if } T > h, v \in W_n, \\ \emptyset, & \text{otherwise,} \end{cases}$$



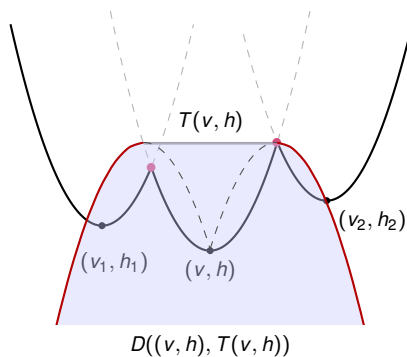
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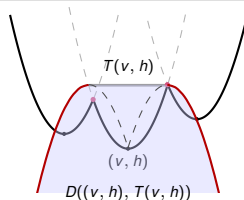
Probability of a *large* region of stabilization :

$$\mathbb{P}((v', h') \in R_n((v, h), \eta + \delta_{(v, h)})) \\ = \mathbb{1}\{v \in W_n\} \mathbb{P}(T > h, (v', h') \in D).$$



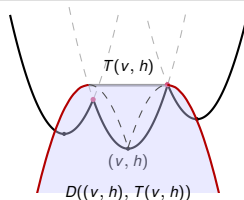
The condition $(v', h') \in D$ implies T must be large :

$$\begin{aligned}(v', h') \in D &\iff \sqrt{T-h} + \sqrt{T-h'} \geq \|v - v'\| \\ &\implies T \geq \|v - v'\|^2/4 + (h \wedge h').\end{aligned}$$



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Tail bounds for T (B., Gusakova '25+)

Let $x = (v, h) \in \mathbb{R}^d \times \mathbb{R}$ and $H \in \mathbb{R}$. Then $\exists C \in (0, \infty)$ such that

$$\mathbb{P}(T(x, \eta_\beta \cup \{x\}) > H) \leq C \exp(-c(H \vee h)^{d/2+\beta+1}), \quad \forall H, h \geq 0,$$

$$\mathbb{P}(T(x, \eta'_\beta \cup \{x\}) > H) \leq C \exp(-c(-(H \vee h))^{-(\beta-d/2-1)}), \quad \forall H, h < 0,$$

$$\mathbb{P}(T(x, \tilde{\eta} \cup \{x\}) > H) \leq C((1-h) \vee 1)^{d/2} \exp(-c e^{(H \vee h)/2}), \quad \forall H, h \in \mathbb{R}.$$

Further Directions

➤ Immediately applicable for functionals of the form

$$F'_n(\eta, w) := \sum_{(v, h) \in \eta} \mathbb{1}(v \in W_n) \mathbb{1}((v, h) \text{ is extreme in } \mathcal{L}(\eta)) w((v, h), \eta),$$

where w is a weight depending on (v, h) and its cell. Still require some moment upper bounds and variance lower bounds.

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- Examples include

(i) **# k -dimensional faces** with centres in W_n :

$$w((v, h), \eta) = \frac{1}{d+1-k} \sum_{F \in \mathcal{F}_k(C((v, h), \eta))} \mathbb{1}(z(F) \in W_n),$$

where $\mathcal{F}_k(P)$ is the set of all k -dimensional faces of P and $z(P)$ is the *center* of P ,

(ii) the **total volume** of all cells of $\mathcal{L}(\eta)$ whose nuclei v fall within W_n .

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- In β' -model, our limitation $\beta > 5d + 1$ is very likely suboptimal. We believe our methods can be pushed a but further to likely achieve the result for $\beta > 3d + 1$.
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Thank you for your attention!